SELF-CONSISTENCY IMPROVES CHAIN OF THOUGHT REASONING IN LANGUAGE MODELS

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ABSTRACT

Chain-of-thought prompting combined with pre-trained large language models has achieved encouraging results on complex reasoning tasks. In this paper, we propose a new decoding strategy, *self-consistency*, to replace the naive greedy decoding used in chain-of-thought prompting. It first samples a diverse set of reasoning paths instead of only taking the greedy one, and then selects the most consistent answer by marginalizing out the sampled reasoning paths. Self-consistency leverages the intuition that a complex reasoning problem typically admits multiple different ways of thinking leading to its unique correct answer. Our extensive empirical evaluation shows that self-consistency boosts the performance of chain-of-thought prompting with a striking margin on a range of popular arithmetic and commonsense reasoning benchmarks, including GSM8K (+17.9%), SVAMP (+11.0%), AQuA (+12.2%), StrategyQA (+6.4%) and ARC-challenge (+3.9%).

1 INTRODUCTION

Although language models have demonstrated remarkable success across a range of NLP tasks, their ability to demonstrate reasoning is often seen as a limitation, which cannot be overcome solely by increasing model scale (Rae et al., 2021; BIG-bench collaboration, 2021, *inter alia*). In an effort to address this shortcoming, Wei et al. (2022) have proposed *chain-of-thought prompting*, where a language model is prompted to generate a series of short sentences that mimic the reasoning process a person might employ in solving a task. For example, given the question *"If there are 3 cars in the parking lot and 2 more cars arrive, how many cars are in the parking lot?"*, instead of directly responding with *"5"*, a language model would be prompted to respond with the entire chain-of-thought: *"There are 3 cars in the parking lot already. 2 more arrive. Now there are 3 + 2 = 5 cars. The answer is 5."*. It has been observed that chain-of-thought prompting significantly improves model performance across a variety of multi-step reasoning tasks (Wei et al., 2022).

In this paper, we introduce a novel decoding strategy called *self-consistency* to replace the greedy decoding strategy used in chain-of-thought prompting (Wei et al., 2022), that further improves language models' reasoning performance by a significant margin. Self-consistency leverages the intuition that complex reasoning tasks typically admit multiple reasoning paths that reach a correct answer (Stanovich & West, 2000). The more that deliberate thinking and analysis is required for a problem (Evans, 2010), the greater the diversity of reasoning paths that can recover the answer.

Figure 1 illustrates the self-consistency method with an example. We first prompt the language model with chain-of-thought prompting, then instead of greedily decoding the optimal reasoning path, we propose a "sample-and-marginalize" decoding procedure: we first *sample* from the language model's decoder to generate a *diverse* set of reasoning paths; each reasoning path might lead to a different final answer, so we determine the optimal answer by *marginalizing out* the sampled reasoning paths to find the most consistent answer in the final answer set. Such an approach is analogous to the human experience that if multiple different ways of thinking lead to the same answer, one has greater confidence that the final answer is correct. Compared to other decoding methods, self-consistency avoids the repetitiveness and local-optimality that plague greedy decoding, while mitigating the stochasticity of a single sampled generation.

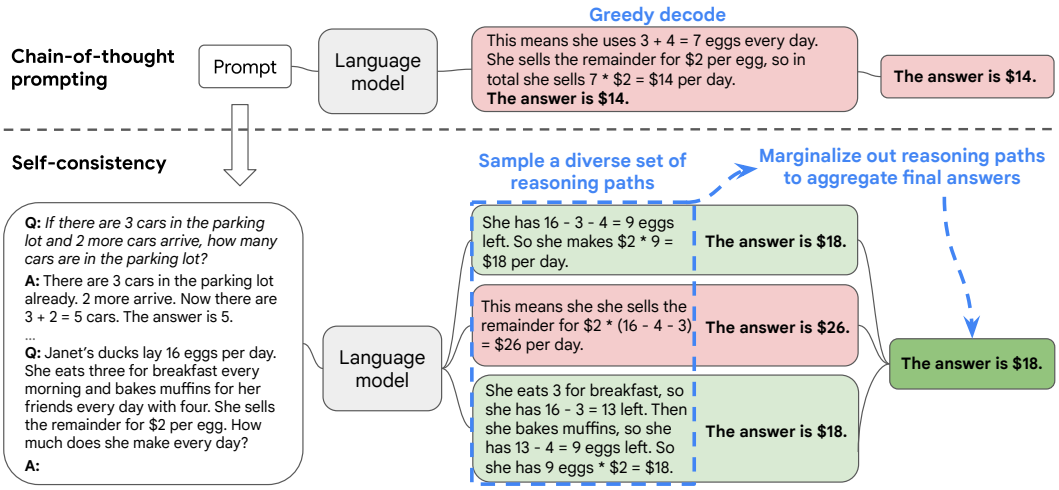

Figure 1: The self-consistency method contains three steps: (1) prompt a language model using chain-of-thought (CoT) prompting; (2) replace the "greedy decode" in CoT prompting by sampling from the language model's decoder to generate a diverse set of reasoning paths; and (3) marginalize out the reasoning paths and aggregate by choosing the most consistent answer in the final answer set.

Self-consistency is far simpler than prior approaches that either train an additional verifier (Cobbe et al., 2021) or train a re-ranker given additional human annotations to improve generation quality (Thoppilan et al., 2022). Instead, self-consistency is entirely *unsupervised*, works off-the-shelf with pre-trained language models, requires no additional human annotation, and avoids any additional training, auxiliary models or fine-tuning. Self-consistency also differs from a typical ensemble approach where multiple models are trained and the outputs from each model are aggregated, it acts more like a "self-ensemble" that works on top of a *single* language model.

We evaluate self-consistency on a wide range of arithmetic and commonsense reasoning tasks over four language models with varying scales: the public UL2-20B (Tay et al., 2022) and GPT-3-175B (Brown et al., 2020), and two densely-activated decoder-only language models: LaMDA-137B (Thoppilan et al., 2022) and PaLM-540B (Chowdhery et al., 2022). On all four language models, self-consistency improves over chain-of-thought prompting by a striking margin across all tasks. In particular, when used with PaLM-540B or GPT-3, self-consistency achieves new state-of-the-art levels of performance across arithmetic reasoning tasks, including GSM8K (Cobbe et al., 2021) (+17.9% absolute accuracy gains), SVAMP (Patel et al., 2021) (+11.0%), AQuA (Ling et al., 2017) (+12.2%), and across commonsense reasoning tasks such as StrategyQA (Geva et al., 2021) (+6.4%) and ARC-challenge (Clark et al., 2018) (+3.9%). In additional experiments, we show self-consistency can robustly boost performance on NLP tasks where adding a chain-of-thought might hurt performance compared to standard prompting (Ye & Durrett, 2022). We also show self-consistency significantly outperforms sample-and-rank, beam search, ensemble-based approaches, and is robust to sampling strategies and imperfect prompts.

2 Self-Consistency over Diverse Reasoning Paths

A salient aspect of humanity is that people think differently. It is natural to suppose that in tasks requiring deliberate thinking, there are likely several ways to attack the problem. We propose that such a process can be simulated in language models via sampling from the language model's decoder. For instance, as shown in Figure 1, a model can generate several plausible responses to a math question that all arrive at the same correct answer (Outputs 1 and 3). Since language models are not perfect reasoners, the model might also produce an incorrect reasoning path or make a mistake in one of the reasoning steps (e.g., in Output 2), but such solutions are less likely to arrive at the *same* answer. That is, we hypothesize that correct reasoning processes, even if they are diverse, tend to have greater agreement in their final answer than incorrect processes.

We leverage this intuition by proposing the following *self-consistency* method. First, a language model is prompted with a set of manually written chain-of-thought exemplars (Wei et al., 2022). Next,

	GSM8K	MultiArith	AQuA	SVAMP	CSQA	ARC-c
Greedy decode	56.5	94.7	35.8	79.0	79.0	85.2
Weighted avg (unnormalized)	56.3 $\pm$ 0.0	90.5 $\pm$ 0.0	35.8 $\pm$ 0.0	73.0 $\pm$ 0.0	74.8 $\pm$ 0.0	82.3 $\pm$ 0.0
Weighted avg (normalized)	22.1 $\pm$ 0.0	59.7 $\pm$ 0.0	15.7 $\pm$ 0.0	40.5 $\pm$ 0.0	52.1 $\pm$ 0.0	51.7 $\pm$ 0.0
Weighted sum (unnormalized)	59.9 $\pm$ 0.0	92.2 $\pm$ 0.0	38.2 $\pm$ 0.0	76.2 $\pm$ 0.0	76.2 $\pm$ 0.0	83.5 $\pm$ 0.0
Weighted sum (normalized)	74.1 $\pm$ 0.0	99.3 $\pm$ 0.0	48.0 $\pm$ 0.0	86.8 $\pm$ 0.0	80.7 $\pm$ 0.0	88.7 $\pm$ 0.0
Unweighted sum (majority vote)	74.4 $\pm$ 0.1	99.3 $\pm$ 0.0	48.3 $\pm$ 0.5	86.6 $\pm$ 0.1	80.7 $\pm$ 0.1	88.7 $\pm$ 0.1

Table 1: Accuracy comparison of different answer aggregation strategies on PaLM-540B.

we sample a set of candidate outputs from the language model's decoder, generating a diverse set of candidate reasoning paths. Self-consistency is compatible with most existing sampling algorithms, including temperature sampling (Ackley et al., 1985; Ficler & Goldberg, 2017), top-k sampling (Fan et al., 2018; Holtzman et al., 2018; Radford et al., 2019), and nucleus sampling (Holtzman et al., 2020). Finally, we aggregate the answers by marginalizing out the sampled reasoning paths and choosing the answer that is the most consistent among the generated answers.

In more detail, assume the generated answers $\mathbf{a}_i$ are from a fixed answer set, $\mathbf{a}_i \in \mathbb{A}$, where $i = 1, \ldots, m$ indexes the m candidate outputs sampled from the decoder. Given a prompt and a question, self-consistency introduces an additional latent variable $\mathbf{r}_i$, which is a sequence of tokens representing the reasoning path in the i-th output, then couples the generation of $(\mathbf{r}_i, \mathbf{a}_i)$ where $\mathbf{r}_i \rightarrow \mathbf{a}_i$, i.e., generating a reasoning path $\mathbf{r}_i$ is optional and only used to reach the final answer $\mathbf{a}_i$. As an example, consider Output 3 from Figure 1: the first few sentences "*She eats 3 for breakfast ... So she has 9 eggs * $2 = $18.*" constitutes $\mathbf{r}_i$, while the answer *18* from the last sentence, "*The answer is $18*", is parsed as $\mathbf{a}_i$.[1] After sampling multiple $(\mathbf{r}_i, \mathbf{a}_i)$ from the model's decoder, self-consistency applies a marginalization over $\mathbf{r}_i$ by taking a majority vote over $\mathbf{a}_i$, i.e., $\arg\max_a \sum_{i=1}^{m} \mathbb{1}(\mathbf{a}_i = a)$, or as we defined as the most "consistent" answer among the final answer set.

In Table 1, we show the test accuracy over a set of reasoning tasks by using different answer aggregation strategies. In addition to majority vote, one can also weight each $(\mathbf{r}_i, \mathbf{a}_i)$ by $P(\mathbf{r}_i, \mathbf{a}_i \mid \text{prompt}, \text{question})$ when aggregating the answers. Note to compute $P(\mathbf{r}_i, \mathbf{a}_i \mid \text{prompt}, \text{question})$, we can either take the unnormalized probability of the model generating $(\mathbf{r}_i, \mathbf{a}_i)$ given $(\text{prompt}, \text{question})$, or we can normalize the conditional probability by the output length (Brown et al., 2020), i.e.,

$$P(\mathbf{r}_i, \mathbf{a}_i \mid \text{prompt}, \text{question}) = \exp^{\frac{1}{K}\sum_{k=1}^{K} \log P(t_k \mid \text{prompt}, \text{question}, t_1, \ldots, t_{k-1})}, \tag{1}$$

where $\log P(t_k \mid \text{prompt}, \text{question}, t_1, \ldots, t_{k-1})$ is the log probability of generating the k-th token t_k in $(\mathbf{r}_i, \mathbf{a}_i)$ conditioned on the previous tokens, and K is the total number of tokens in $(\mathbf{r}_i, \mathbf{a}_i)$. In Table 1, we show that taking the "unweighted sum", i.e., taking a majority vote directly over $\mathbf{a}_i$ yields a very similar accuracy as aggregating using the "normalized weighted sum". We took a closer look at the model's output probabilities and found this is because for each $(\mathbf{r}_i, \mathbf{a}_i)$, the normalized conditional probabilities $P(\mathbf{r}_i, \mathbf{a}_i \mid \text{prompt}, \text{question})$ are quite close to each other, i.e., the language model regards those generations as "similarly likely".[2] Additionally, when aggregating the answers, the results in Table 1 show that the "normalized" weighted sum (i.e., Equation 1) yields a much higher accuracy compared to its unnormalized counterpart. For completeness, in Table 1 we also report the results by taking a "weighted average", i.e., each a gets a score of its weighted sum divided by $\sum_{i=1}^{m} \mathbb{1}(\mathbf{a}_i = a)$, which results in a much worse performance.

Self-consistency explores an interesting space between open-ended text generation and optimal text generation with a fixed answer. Reasoning tasks typically have fixed answers, which is why researchers have generally considered greedy decoding approaches (Radford et al., 2019; Wei et al., 2022; Chowdhery et al., 2022). However, we have found that even when the desired answer is fixed, introducing diversity in the reasoning processes can be highly beneficial; therefore we leverage

[1] The parser is task dependent. For arithmetic reasoning, we parse the first numerical part as the final answer after the model generates "The answer is ". For commonsense reasoning, we parse the full string answer as the final answer after the model generates "The answer is ". Most generated outputs have a consistent format of "{Reasoning paths}. The answer is X." if we prompt the language model in this format.

[2] This also means that the language model is not well calibrated and thus cannot distinguish well between correct solutions and wrong solutions, which also explains why additional re-rankers were trained to better judge the quality of the solutions in previous work (Cobbe et al., 2021; Thoppilan et al., 2022).